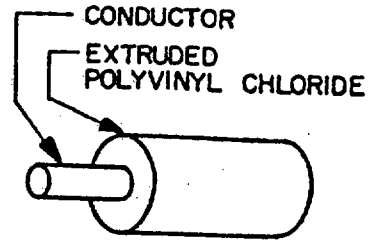
MIL-W-16878/2A(NAVY)
10 August 1981
SUPERSEDING
MIL-W-16878/2(NAVY)(IN PART)
16 January 1961
(See supersession data herein)

MILITARY SPECIFICATION SHEET

WIRE, ELECTRICAL, POLYVINYL CHLORIDE (PVC) INSULATED, 105°C, 1000 VOLTS

This specification sheet is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the wire described herein shall consist of this document and the latest issue of MIL-W-16878.



SH11874

FIGURE 1. Construction.

TABLE I. Construction details.

Wire size	Stranding	Conductor		Conductor diameter (nominal) (inch)	Finished wire	
		Material 2/ 3/	Coating		Diameter (inch)	
					Min	Max
26	1 X 26	Copper	Tin	0.0159	0.046	0.054
26	1 X 26	H.S.C.A.	Silver	.0159	.046	.054
26	1 X 26	C.C. Steel	Tin	.0159	.046	.054
26	7 X 34	Copper	Tin	.019	.049	.057
26	7 X 34	H.S.C.A.	Silver	.019	.049	.057
26	19 X 38	Copper	Tin	.021	.049	.057
26	19 X 38	H.S.C.A.	Silver	.021	.049	.057

See footnotes at end of table.

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TABLE I. Construction details.- Continued

Wire size	Stranding	Conductor		Conductor diameter (nominal) (inch)	Finished wire	
		Material 2/ 3/	Coating		Diameter (inch)	
					Min	Max
24	1 X 24	Copper	Tin	0.0201	0.050	0.058
1/ 24	1 X 24	H.S.C.A.	Silver	.0201	.050	.058
1/ 24	1 X 24	C.C. Steel	Tin	.0201	.050	.058
24	7 X 32	Copper	Tin	.024	.054	.062
24	7 X 32	H.S.C.A.	Silver	.024	.054	.062
24	19 X 36	Copper	Tin	.026	.054	.062
24	19 X 36	H.S.C.A.	Silver	.026	.054	.062
22	1 X 22	Copper	Tin	.0254	.055	.064
1/ 22	1 X 22	H.S.C.A.	Silver	.0254	.055	.064
1/ 22	1 X 22	C.C. Steel	Tin	.0254	.055	.064
22	7 X 30	Copper	Tin	.030	.060	.068
22	7 X 30	H.S.C.A.	Silver	.030	.060	.068
22	19 X 34	Copper	Tin	.033	.060	.068
22	19 X 34	H.S.C.A.	Silver	.033	.060	.068
20	1 X 20	Copper	Tin	.0320	.062	.070
20	1 X 20	H.S.C.A.	Silver	.0320	.062	.070
20	7 X 28	Copper	Tin	.038	.068	.076
20	7 X 28	H.S.C.A.	Silver	.038	.068	.076
20	10 X 30	Copper	Tin	.038	.068	.076
20	10 X 30	H.S.C.A.	Silver	.038	.068	.076
20	19 X 32	Copper	Tin	.041	.068	.076
20	19 X 32	H.S.C.A.	Silver	.041	.068	.076
18	1 X 18	Copper	Tin	.0403	.070	.079
18	7 X 26	Copper	Tin	.049	.079	.087
18	16 X 30	Copper	Tin	.047	.077	.085
18	19 X 30	Copper	Tin	.051	.079	.087
16	1 X 16	Copper	Tin	.0508	.080	.089
16	19 X 29	Copper	Tin	.059	.089	.097
16	26 X 30	Copper	Tin	.062	.092	.100
14	1 X 14	Copper	Tin	.0641	.094	.102
14	19 X 27	Copper	Tin	.072	.102	.110
1/ 14	41 X 30	Copper	Tin	.080	.110	.118
1/ 12	1 X 12	Copper	Tin	.0808	.110	.119
12	19 X 25	Copper	Tin	.091	.121	.129
12	37 X 28	Copper	Tin	.089	.119	.127
12	65 X 30	Copper	Tin	.098	.128	.136

1/ Inactive for new design.
2/ H.S.C.A. stands for high strength copper alloy.
3/ C.C. Steel stands for copper clad steel.

REQUIREMENTS:
Visual and mechanical inspection: Applicable.
Spark test: 5.0 kilovolts (kV)
Impulse dielectric test: 10.0 kV.

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Tank test: 3.0 kV.
Dielectric withstanding voltage: 3.0 kV.
Insulation resistance: $IR = K \log_{10} D/d$.

Where: IR = Minimum insulation resistance in megohms per 1000 feet at 20°C.
K = 2000
D = Maximum average diameter of finished wire.
d = Conductor diameter.

Cold bend: Condition 4 hours at minus 54°C ± 1°C (see table II).

TABLE II. Cold bend mandrel sizes.

Wire size	Cold bend mandrel (maximum diameter) (inches)
26 through 16	2
14, 12	3

Concentricity: 70 percent (minimum).
Surface resistance: 5 megohms (minimum).
Wrap back: Not applicable.
Heat resistance: Condition at 150°C.
Flammability: Applicable - 30 seconds (maximum) burning time
3 inches (maximum) flame travel
Soldering: Applicable.
Heat aging: 25 percent change (maximum) 96 hours at 136 C.°
Insulation tensile strength: 1800 pounds force per square inch (minimum).
Insulation elongation: 100 percent (minimum).
Stripe durability: Applicable.
Fungus resistance: Applicable.
Maximum direct current resistance of finished wire: See table I of MIL-W-16878.

Military part number (see MIL-W-16878):

M16878/ 2 B C B 905

Military designation
Specification sheet
Conductor material
Conductor size
Conductor stranding
Color code

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Supersession data: The wire by this specification sheet supersedes MIL-W-16878/2(NAVY), type C, without jacket.

Review activities:
Navy - AS, EC

User activities:
Navy - OS, MC

Preparing activity:
Navy - SH
(Project 6145-N300-2)

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